INCH-POUND

MIL-F-63446B (AR)
12 September 1994
SUPERSEDING
MIL-F-63446A (AR)
12 September 1983

MILITARY SPECIFICATION

FUZES, GENERAL PURPOSE GRENADE: M230 AND M231 (PRACTICE)

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1. Scope This specification covers the requirements, examinations, and tests for the M230 and M231 Fuzes with detonator, used with the M73 GP Grenade and M75 GP Grenade, respectively. It covers quality assurance provisions and special test requirements not covered by the drawings for parts, assembly and packaging of the fuzes. (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, stanards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-D-14978	-	Detonator, Stab, M55: Loading, Assembly and Packing.
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specifications for
MIL-A-70625	-	Automated Acceptance Inspection Equipment for Non-Electronic Components, Design, Testing and Approval, of

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey, 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:27Z
Check the source to verify that this is the current version before use.

MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Tests for
MIL-STD-810	-	Environmental Test Methods
MIL-STD-1167	-	Ammunition Data Cards

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.5)

US ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8799112	-	Marking Diagram and Sealing for Interplant or CONUS Shipment of Ammunition Fiberboard Boxes
9333825	-	Fuze General Purpose Grenade: M230
9340491	-	Box, Fiber, Inner, Packing, Ammunition for Fuze, M230/M231
9340492	-	Box, Fiber, Outer, Packing Ammunition for Fuze, M230/M231
9345168	-	Fuze General Purpose Grenade Practice: M231

(Copies of other documents, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the document which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B85	-	Standard Specification for Aluminum-Alloy Die Castings

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria.)

3. REQUIREMENTS

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:27Z
Check the source to verify that this is the current version before use.

3.1 Materials. Materials shall be as specified herein and on the applicable drawings and specifications.

3.2 First article When specified in the contractor purchase order (see 6.2), a sample shall be subjected to fist article in accordance with the technical provisions herein (see 4.3).

3.3 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the fuze. Any data (e.g., shop drawings, layouts flowsheets, processing procedures, etc.) prepared by the contractor or obtained from a vender to support fabrication and manufacture of the production item shall be made available, upon request to, for inspection by the contracting officer or his designated representative. The dimensions shown on the drawings apply at room temperatures from 68°F to 70°F.

3.4 Fuze assemblies and parts Fuze assemblies and parts shall comply with all the requirements specified on Drawings 9333825, 9345168, associated drawings as applicable, and with all requirements specified in applicable specifications and standards.

3.4.1 Molded plastic parts Dimensions on plastic parts that area function of core and cavity sizes of the mold itself shall be inspected at reduced levels. One tenth of one percent (.1%) of a production lot shall be inspected per cavity. A single reject resulting from the reduced inspection of molded dimensions shall cause that cavity to be removed from production until corrective action is taken and the part from that cavity requalified.

3.4.2 Die cast parts The aluminum die cast fuze body shall be manufactured, tested and inspected per the requirements of ASTM B85 and Drawing 9333824.

3.4.3 Porosity check. The die cast fuze body shall be inspected for porosity in accordance with 4.5.1 of this specification.

3.4.4 Springs.

3.4.4.1 Spring embrittlement. The springs used in the fabrication of the fuze shall not break when extended or distorted to the point of minimum permanent distortion.

3.4.4.2 Spring set. The slider spring and firing pin spring shall maintain drawing load requirements after being compressed at elevated temperatures.

3.4.5 Detonator. All detonators shall have been subjected to and passed all the requirements of MIL-D-14978.

3.4.6 Lead output. The long and short lead assemblies, when initiated in tandem by an M55 detonator, shall cause an impression in a steel witness block not less than .015 inch in depth.

3.5 Functional requirements.

3.5.1 Jolt. The fuze shall be safe to store, handle, use or dispose of after being subjected to Test A1, Jolt Machine Method of MIL-STD-331.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

3.5.2 Jumble. The fuze shall be safe to store, handle, use or dispose of after being subjected to Test A2, of MIL-STD-331.

3.5.3 Transportation vibration. The fuze shall be safe and meet all functional requirements of this specification after being subjected to B1, Section 1-5 of MIL-STD-331.

3.5.4 Helicopter vibration. The fuze shall be safe and meet all functional requirements of this specification after being subjected to Method 514 of MIL-STD-810.

3.5.5 Five foot drop. The fuze shall be safe to store, handle, use or dispose of after being subjected to Test A4, Single Drop Procedure of MIL-STD-331. The fuze need not be operable.

3.5.6 Thermal shock. The fuze shall be safe and meet all functional requirements of this specification after being subjected to Test C7 of MIL-STD-331.

3.5.7 Shear wire test. It shall require 18.75 ± 3.25 pounds of force to sever the shear wire of an assembled fuze when a constant pull rate is applied in a direction within two degrees parallel to the axis of the arming pin.

3.5.8 Safety pin removal. It shall require not less than one pound and not more than eight pounds of force to remove the safety pin from the fuze when the direction of the applied load is within five degrees of the axis of the safety pin. The load shall be applied at a constant rate.

3.5.9 Arming time After removal of the safety pin and upon withdrawal of the arming pin, the slider shall travel the full arming stroke in an elapsed time of 1.50 ± .50 seconds.

3.5.10 Fuzing no-go. An armed fuze shall not trigger (release firing pin) when subjected to the No-Go Shock Test.

3.5.11 Fuzing go. An armed fuze shall trigger and initiate the detonator when subjected to the Go Shock Test

3.5.12 Arming pin/slider lock. The arming pin shall lock the slider in its fully safe position when the fuze is fully assembled.

3.5.13 Auto trigger. The trigger shall not function spontaneously when the slider is moved to the fully armed position.

3.5.14 Temperature. The fuze shall withstand exposure to and function after a storage temperature of -50°F ± 5°F, 70°F ± 15°F and 160°F ± 5°F.

3.6 Data requirements. The contractor shall generate data in accordance with the requirements of 4.5.18.

3.7 Workmanship. All parts and accessories shall be free from burrs, chips, sharp edges, unblended radii, cracks, surface checks, blisters, excess flash, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing the base material. The loading and assembly operations shall be performed in a thorough workmanlike manner consistent with the best practices of modem industry. All required markings shall be neat and sharply defined.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contractor purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contractor purchase order, the contractor may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapprove by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as a part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The following types of inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a frost article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

Part Description	Drawing	Quantity
Slider	9333807	25/Cavity
Body, Fuze	9333824	25/Cavity
Spring, Slider	9333811	25
Screw, Pivot	9333808	25
Anti-twist Pin	9347220	25
Sleeve, Fuze Body	9345131	25
Fuze Body Assembly	9347185	25
Trigger	9333812	25
Pin, Arming	9333814	25
wire, shear	9345128	25
Slider, M55 Detonator Assembly	9347219	25
Pin, Safety	9333818	25
Pinion, Escapement	8833366	25
Gear Escapement	8833365	25
Escapement Gear-Pinion Assembly	8833364	25
Pinion No. 1	8833360	25
Gear No. 1	8833359	25
No. 1 Gear-Pinion Assembly	8833358	25
Pinion No. 2	8833363	25
Gear No. 2	8833362	25
No. 2 Gear-Pinion Assembly	8833361	25
Lever	8833369	25
Pallet, Lever	8833368	25
Shaft, Lever	8833370	25
Lever Assembly	8833367	25
Plate, Bottom	8833356	25/die
Plate, Top	8833357	25/die
Pillar	8833355	25
Escapement Gear Assembly	8833353	50
Body, Firing Pin	9333816	25
Pin, Straight	9345321	25
Firing Pin, Subassembly	9345130	50
Cap, Firing Pin	9333817	25
Spring, Firing Pin	9333810	25
Firing Pin Assembly	9347182	25
Ball, Detent	9349223	25
Plate, Barrier	9333809	25
Spring, Arming Pin Lock	9345129	25
Disc, Closing	9333821	25
Lead Cup Short	9333822	50
Lead Cup Assembly, Short	9333823	50
Lead Cup, Long	9333819	25
Lead Cup Assembly, Long	9333820	50
Plate Assembly, Barrier	9345166	50
Drive Screw, Modified	9390620	25
Barrier Plate Assembly with Retaining Tape	12561493	50
Fuze, General Purpose Grenade Practice: M231	9345168	239(min)
Fuze, General Purpose Grenade: M230	9333825	239(min)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. E. rticle inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
CLASSIFICATION	Fuzes, General Purpose Grenade: M230 and M231	1 OF 11	See Below

EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
Slider			
(Dwg. 9333807)			
Examination for defects	25/Cavity	3.4.1	4.4.2.1
Body, Fuze			
(Dwg. 9333824)			
Porosity in firing pin bore	25/Cavity	3.4.3	4.5.1
Examination for defects	25/Cavity	3.4.2	4.4.2.2
Spring, Slider			
(Dwg. 93338111)			
Spring Load Requirement	25	3.4	Gage
Spring Embrittlement	25	3.4.4.1	4.5.2
Spring Set	25	3.4.4.2	4.5.3
Examination for defects	25	3.4	4.4.2.3
Screw, Pivot			
(Dwg. 9333808)			
Examination for defects	25	3.4	4.4.2.4
Anti-Twist Pin			
(Dwg. 9347220)			
Examination for defects	25	3.4	4.4.2.5

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Fuzes, General Purpose Grenade: M230 and M231	2 OF 11	See Below

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
	Sleeve, Fuze Body			
	(Dwg. 9345131)			
	Examination for defects	25	3.4	4.4.2.6
	Fuze Body Assembly			
	(Dwg. 9347185)			
	Anti-twist pushout requirement	25	3.4	Gage
	Sleeve pushout requirement	25	3.4	Gage
	Examination for defects	25	3.4	4.4.2.7
	Trigger			
	(Dwg. 9333812)			
	Examination for defects	25	3.4	4.4.2.8
	Pin, Arming			
	(Dwg. 9333814)			
	Examination for defects	25	3.4	4.4.2.9

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. Fi rticle inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
CLASSIFICATION	Fuzes, General Purpose Grenade: M230 and M231	3 OF 11	See Below

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
	Wire, Shear			
	(Dwg. 9345128)			
	Examination for defects	25	3.4	4.4.2.10
	Shear load	25	3.4	Gage
	Slider, M55 Detonator Assembly			
	(Dwg. 9347219)			
	Examination for defects	25	3.4	4.4.2.11
	Detonator Deterioration	25	3.4.5	4.5.4
	Pin, Safety			
	(Dwg. 9333818)			
	Examination for Defects	25	3.4	4.4.2.12
	Pinion, Escapement			
	(Dwg. 8833366)			
	Examination for Defects	25	3.4	4.4.2.13
	Gear Escapement			
	(Dwg. 8833365)			
	Examination for Defects	25	3.4	4.4.2.14

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Fuzes, General Purpose Grenade: M230 and M231	SHEET 4 OF 11	DRAWING NUMBER See Below
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH / NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
	Escapement Gear-Pinion Assembly (Dwg. 8833364) Examination for Defects	25	3.4 / 4.4.2.15
	Pinion No. 1 (Dwg. 8833360) Examination for Defects	25	3.4 / 4.4.2.16
	Gear No. 1 (Dwg. 8833359) Examination for Defects	25	3.4 / 4.4.2.17
	No. 1 Gear-Pinion Assembly (Dwg. 8833358) Examination for Defects	25	3.4 / 4.4.2.18
	Pinion No. 2 (Dwg. 8833363) Examination for Defects	25	3.4 / 4.4.2.19
	Gear No. 2 (Dwg. 8833362) Examination for Defects	25	3.4 / 4.4.2.20

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. F article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Fuzes, General Purpose Grenade: M230 and M231	5 OF 11	See Below

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
	No. 2 Gear-Pinion Assembly (Dwg. 8833361) Examination for Defects	25	3.4	4.4.2.21
	Lever (Dwg. 8833369) Examination for Defects	25	3.4	4.4.2.22
	Pallet, Lever (Dwg. 8833368) Examination for Defects	25	3.4	4.4.2.23
	Shaft, Lever (Dwg. 8833370) Examination for Defects	25	3.4	4.4.2.24
	Lever Assembly (Dwg. 8833367) Examination for Defects	25	3.4	4.4.2.25
	Plate, Bottom (Dwg. 8833356) Examination for Defects	25/die	3.4	4.4.2.26

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE Fuzes, General Purpose Grenade: M230 and M231	SHEET 6 OF 11	DRAWING NUMBER See Below
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH / NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
	Plate, Top (Dwg. 8833357) Examination for Defects	25/die	3.4 4.4.2.27
	Pillar (Dwg. 8833355) Examination for Defects	25	3.4 4.4.2.28
	Escapement Gear Assembly (Dwg. 8833353) Axial Pull of Assembly Examination for Defects	50 50	3.4 Gage 3.4 4.4.2.29
	Body, Firing Pin (Dwg. 9333816) Examination for Defects	25	3.4 4.4.2.30
	Pin Straight (Dwg. 9345321) Examination for Defects	25	3.4 4.4.2.31
	Firing Pin, Subassembly (Dwg. 9345130) Examination for Defects	50	3.4 4.4.2.32

NOTES.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. F article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH CLASSIFICATION	TITLE / EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	DRAWING NUMBER / NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
	Fuzes, General Purpose Grenade: M230 and M231 — SHEET 7 OF 11			See Below
	Cap, Firing Pin			
	(Dwg. 9333817)			
	Examination for Defects	25	3.4	4.4.2.33
	Spring, Firing Pin			
	(Dwg. 9333810)			
	Spring Embrittlement	25	3.4.4.1	4.5.2
	Spring Set	25	3.4.4.2	4.5.3
	Examination for Defects	50	3.4	4.4.2.34
	Firing Pin Assembly			
	(Dwg. 9347182)			
	Examination for Defects	25	3.4	4.4.2.35
	Ball, Detent			
	(Dwg. 9349223)			
	Examination for Defects	25	3.4	4.4.2.36
	Plate, Barrier			
	(Dwg. 9333809)			
	Examination for Defects	25	3.4	4.4.2.37

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
CLASSIFICATION	Fuzes, General Purpose Grenade: M230 and M231	8 OF 11	See Below

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
	Spring, Arming Pin Lock			
	(Dwg. 9345129)			
	Examination for Defects	25	3.4	4.4.2.38
	Disc Closing			
	(Dwg. 9333821)			
	Examination for Defects	25	3.4	4.4.2.39
	Lead Cup, Short			
	(Dwg. 9333822)			
	Examination for Defects	50	3.4	4.4.2.40
	Lead Cup Assembly, Short			
	(Dwg. 9333823)			
	Lead Output	50	3.4.6	4.5.5
	Examination for Defects	50	3.4	4.4.2.41
	Lead Cup, Long			
	(Dwg. 9333819)			
	Examination for Defects	25	3.4	4.4.2.42
	Lead Cup Assembly, Long			
	(Dwg. 9333820)			
	Lead Output	50	3.4.6	4.5.5
	Examination for Defects	50	3.4	4.4.2.43

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. F article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE Fuzes, General Purpose Grenade: M231	SHEET 9 OF 11	DRAWING NUMBER 9345168
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH / NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
	Plate Assembly, Barrier (Dwg. 9345166) Examination for Defects	25	3.4 / 4.4.2.44
	Drive Screw, Modified (Dwg. 9390620) Examination for Defects	25	3.4 / 4.4.2.45
	Barrier Plate Assembly with Retainer Tape (Dwg. 12561493) Examination for Defects	50	3.4 / 4.4.2.46

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH CLASSIFICATION	TITLE Fuzes, General Purpose Grenade: M231	SHEET 10 OF 11	DRAWING NUMBER 9345168
			NEXT HIGHER ASSEMBLY

EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Jolt	12	3.5.1	4.5.6
Jumble	12	3.5.2	4.5.7
Transportation Vibration	20	3.5.3	4.5.8/4.5.15(a)/4.5.16(a)
Helicopter Vibration	30	3.5.4	4.5.9/4.5.15(a)/4.5.16(a)
Five Foot Drop	20	3.5.5	4.5.10
Thermal Shock	30	3.5.6	4.5.11/4.5.15(a)/4.5.16(a)
Shear Wire Pull Test	115 (b)	3.5.7	4.5.12
Safety Pin Pull Test	115 (b)	3.5.8	4.5.13
Arming Time	115 (b)	3.5.9	4.5.14(c)
Fuzing No-Go	49 (b)	3.5.10	4.5.15(d)
Fuzing Go	66 (b)	3.5.11	4.5.16(e)
Arming Pin/Slider Lock	239 min (f)	3.5.12	4.5.17
Auto Trigger	239 min (f)	3.5.13	4.5.17
Torque Requirement	115(b)	3.4	4.4.2.47
Examination for Defects	239 min (f)	3.4	4.4.2.47

NOTES:

(a) Fuzing No-Go - 1/2 test sample at -50° F ± 5° F, 1/2 test sample at 160° F ± 5° F; Fuzing Go - 1/2 test sample at -50° F ± 5° F, 1/2 test sample at 160° F ± 5° F.

(b) The same fuzes are to be subjected to these tests. If the test within 90 seconds requirement cannot be maintained, however, each test or any combination of these tests maybe performed on a separate group of fuzes which could increase the maximum sample size.

(c) Prior to performing the test, condition fuzes for two hours after stabilization at indicated temperatures: -50° F ± 5° F -38 fuzes; 70° F ± 15° F -39 fuzes; 150° F ± 5° F -38 fuzes. Test shall be completed within 90 seconds of removal from temperature conditioning chambers. All fuzes tested at cold temperature shall be conditioned while sealed in plastic bags. (d) Prior to performing the test, condition fuzes for two hours after stabilization at indicated temperatures: -50° F ± 5° F -16 fuzes; 70° F ± 15° F -17 fuzes; 150° F ± 5° F -16 fuzes. Test shall be completed within 90 seconds of removal from temperature conditioning chambers. All fuzes tested at cold temperature shall be conditioned while sealed in plastic bags.

(e) Prior to performing the test, condition fuzes for two hours after stabilization at indicated temperatures: -50° F ± 5° F -22 fuzes; 70° F ± 15° F -22 fuzes; 150° F ± 5° F -22 fuzes. Test shall be completed within 90 seconds of removal from temperature conditioning chambers. All fuzes tested at cold temperature shall be conditioned while sealed in plastic bags.

(f) Actual number of fuzes to be examined for defects will depend upon actual number of fuzes submitted for testing as determined by note (b) above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE I. Fi rticle inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE		DRAWING NUMBER
	Fuzes, General Purpose Grenade: M230	SHEET 11 OF 11	9333825

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
	Jolt	12	3.5.1	4.5.6
	Jumble	12	3.5.2	4.5.7
	Transportation Vibration	20	3.5.3	4.5.8/4.5.15(a)/4.5.16(c)
	Helicopter Vibration	30	3.5.4	4.5.9/4.5.15(a)/4.5.16(a)
	Five Foot Drop	20	3.5.5	4.5.10
	Thermal Shock	30	3.5.6	4.5.11/4.5.15(a)/4.5.16(a)
	Shear Wire Pull Test	115 (b)	3.5.7	4.5.12
	Safety Pin Pull Test	115 (b)	3.5.8	4.5.13
	Arming Time	115 (b)	3.5.9	4.5.14(c)
	Fuzing No-Go	49 (b)	3.5.10	4.5.15(d)
	Fuzing Go	66 (b)	3.5.11	4.5.16(e)
	Arming Pin/Slider Lock	239 min (f)	3.5.12	4.5.17
	Auto Trigger	239 min (f)	3.5.13	4.5.17
	Torque Requirement	115 (b)	3.4	4.4.2.48
	Examination for Defects	239 min(f)	3.4	4.4.2.48

NOTES:

(a) Fuzing No-Go - 1/2 test sample at -50° F ± 5° F, 1/2 test sample at 160° F ± 5° F; Fuzing Go - 1/2 test sample at -50° F ± 5° F, 1/2 test sample at 160° F ± 5° F.

(b) The same fuzes are to be subjected to these tests. If the test within 90 seconds requirement cannot be maintained, however, each test or any combination of these tests maybe performed on a separate group of fuzes which could increase the maximum sample size.

(c) Prior to performing the test, condition fuzes for two hours after stabilization at indicated temperatures: -50° F ± 5° F -38 fuzes; 70° F ± 15° F -39 fuzes; 150° F ± 5° F -38 fuzes. Test shall be completed within 90 seconds of removal from temperature conditioning chambers. All fuzes tested at cold temperature shall be conditioned while sealed in plastic bags.

(d) Prior to performing the test, condition fuzes for two hours after stabilization at indicated temperatures: -50° F ± 5° F -16 fuzes; 70° F ± 15° F -17 fuzes; 150° F ± 5° F -16 fuzes. Test shall be completed within 90 seconds of removal from temperature conditioning chambers. All fuzes tested at cold temperature shall be conditioned while sealed in plastic bags.

(e) Prio to performing the test, condition fuzes for two hours after stabilization at indicated temperatures: -50° F ± 5° F -22 fuzes; 70° F ± 15° F -22 fuzes; 150° F ± 5° F -22 fuzes. Test shall be completed within 90 seconds of removal from temperature conditioning chambers. All fuzes tested at cold temperature shall be conditioned while sealed in plastic bags.

(f) Actual number of fuzes to be examined for defects will depend upon actual number of fuzes submitted for testing as determined by note (b) above.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.3.2 Inspections to be performed. As determined by the Goverment the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (see TABLE I) and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term “inspection lot” is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of loaded fuzes shall contain the following:

a. Detonators from not more than two M55 production lots with same interfix number from one manufacturer.

b. Short and long lead assemblies from not more than two production lots with same interfix number from one manufacturer.

The quantity in each production lot of fuzes shall not exceed 45,000.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor’s quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE II. Attributes sampling inspection.

LOT SIZE			INSPECTION LEVELS						
			I	II	III	IV	V	VI	S-4
2	to	8	*	*	*	*	*	5	*
9	to	15	*	*	*	*	13	5	*
16	to	25	*	*	*	20	13	5	*
26	to	50	*	*	32	20	13	5	32
51	to	90	*	*	32	20	13	7	32
91	to	150	*	125	32	20	13	11	32
151	to	280	*	125	32	20	20	13	32
281	to	500	*	125	48	47	29	16	32
501	to	1200	*	125	73	47	34	19	32
1201	to	3200	1250	125	73	53	42	23	32
3201	to	10000	1250	192	86	68	50	29	32
10001	to	35000	1250	294	108	77	60	35	32
35001	to	150000	1250	294	123	96	74	40	73

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc. may be used by the contractor when they provide as a minimum, the level of quality assurance required by the provisions cited herein. Prior to applying such alternative procedures, methods or equipment the contractor shall, describe them in a written proposal submitted to the Government for evaluation (see 6.6). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

c. Sampling plans for fuze casting. Machined dimensions shall be inspected in accordance with Table II above, using the inspection levels stated in the Classification of Characteristics paragraphs. The sample size for cast dimensions shall be reduced and shall be equal to a minimum of 0.1 % of a production lot per cavity. For cast sampling, accept/reject is 0/ 1 respectively. inspection of the firing pin bore for porosity shall be 100%. For inspection of porosity at locations other than the firing pin bore, use the S-4 inspection level of Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH 4.4.2.1	TITLE Slider	SHEET 1 OF 2	DRAWING NUMBER 9333807
			NEXT HIGHER ASSEMBLY 9333825/9345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Special (b)				
1	Presence of red lacquer on bore rider	100%	3.4	AAIE (c)
Major				
101	Height of triggering mass tab	Level III(a)	3.4	Gage
102	Width of slider rear	Level III(a)	3.4	Gage
103	Thickness of slider rear	Level III(a)	3.4	Gage
104	Height of cage hole	Level III(a)	3.4	Gage
105	Location of cage hole from side of slider	Level III(a)	3.4	Gage
106	Diameter of cage hole	Level III(a)	3.4	Gage
107	Location of detonator hole from bottom	Level III(a)	3.4	Gage
108	Depth of detonator hole	Level III(a)	3.4	Gage
109	Distance from slider side to end of rack	Level III(a)	3.4	Gage
110	Location of arming pin hole from front	Level III(a)	3.4	Gage
111	Location of safety wire hole from front	Level III(a)	3.4	Gage
112	Location of safety wire hole from bottom	Level III(a)	3.4	Gage
113	Gear teeth parameters	Level III(a)	3.4	Gage
114	Radius at cage/body interface	Level III(a)	3.4	Gage

NOTES:
(a) See 3.4.1 for applicability of reduced inspection.
(b) See MIL-A-48078 for definition.
(c) Automated Acceptance Inspection Equipment in accordance with MIL-A-70625.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.1	TITLE Slider	SHEET 2 OF 2	DRAWING NUMBER 9333807
			NEXT HIGHER ASSEMBLY 9333825/9345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
115	Perpendicularity of triggering mass tab to slider body	Level III(a)	3.4	Gage
116	Straightness to slider	Level III(a)	3.4	Gage
117	Minimum coverage of red lacquer bore rider	Level III(a)	3.4	Gage
118	Width of bore rider tip	Level III(a)	3.4	Gage
119	Thickness of slider front	Level III(a)	3.4	Gage
Minor				
201	Diameter of safety pin hole	Level IV(a)	3.4	Gage
202	Diameter of detonator hole	Level IV(a)	3.4	Gage
203	Thickness of bore rider	Level IV(a)	3.4	Gage
204	Thickness of detonator slot	Level IV(a)	3.4	Gage
205	Thickness of cage slot	Level IV(a)	3.4	Gage
206	Location of detonator slot	Level IV(a)	3.4	Gage
207	Length of bore rider	Level IV(a)	3.4	Gage
208	Maximum coverage of red lacquer on bore rider	Level IV(a)	3.4	Gage
209	Width of slider between ends	Level IV(a)	3.4	Gage
210	Thickness of slider between end	Level IV(a)	3.4	Gage
211	Evidence of poor workmanship	Level IV(a)	3.7	Gage

NOTES:
(a). See paragraph 3.4.1 for applicability of reduced inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.2	TITLE Body, Fuze	SHEET 1 OF 3	DRAWING NUMBER 9333824
			NEXT HIGHER ASSEMBLY 9347183/9347185

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Porosity in firing pin hole	100%	3.4.3	4.5.1
102	Diameter of firing pin cap counterbore	Level III (a)	3.4.2	Gage
103	Width of slider slot	Level III (a)	3.4.2	Gage
104	Depth of slider slot	Level III (a)	3.4.2	Gage
105	Location of front end of slider slot	Level III (a)	3.4.2	Gage
106	Location of side of slider slot	Level III (a)	3.4.2	Gage
107	Depth of cage pocket	Level III (a)	3.4.2	Gage
108	Flange diameter	Level III (a)	3.4.2	Gage
109	True position of pivot screw hole to detent ball hole	Level III	3.4.2	Gage
110	Counterbore diameter, arming pin	Level III	3.4.2	Gage
111	True position, center hole to arming pin counterbore	Level III (a)	3.4.2	Gage
112	Radii at bottom of slider slot	Level III (a)	3.4.2	Gage
113	True position of detent ball hole from center side of slot	Level III	3.4.2	Gage
114	Location of timer holes from side of slider slot	Level III	3.4.2	Gage
115	Diameter of drive screw holes	Level III	3.4.2	Gage
116	Location of pivot screw surface from center of firing pin hole	Level III	3.4.2	Gage

NOTES:
(a) Reduced sampling for cast dimensions see 4.4.2 c.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Body, Fuze	2 OF 3	9333824

NEXT HIGHER ASSEMBLY: 9347183/9347185

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Height of firing pin bore	Level IV (a)	3.4.2	Gage
202	Width of key	Level IV (a)	3.4.2	Gage
203	Diameter of hub	Level IV (a)	3.4.2	Gage
204	Overall height	Level IV (a)	3.4.2	Gage
205	Height of intermediate diameter	Level IV (a)	3.4.2	Gage
206	Diameter of firing pin bore at slider slot	Level IV (a)	3.4.2	Gage
207	Location of safety pin wire hole from firing pin bore	Level IV	3.4.2	Gage
208	Location of safety pin wire hole from bottom of body	Level IV	3.4.2	Gage
209	Depth of bore rider slot	Level IV (a)	3.4.2	Gage
211	Diameter RSP holes	Level IV	3.4.2	Gage
212	Max. depth of relief surface	Level IV (a)	3.4.2	Gage
213	Diameter, center hole	Level IV (a)	3.4.2	Gage
214	Diameter of anti-twist pin holes	Level IV (a)	3.4.2	Gage
215	Distance between anti-twist pin holes	Level IV (a)	3.4.2	Gage
216	Diameter of detent ball hole	Level IV (a)	3.4.2	Gage

NOTES:
(a) Reduced sampling for cast dimensions see 4.4.2 c.

Source: https://assist.dla.mil -- Downloaded: 2026-0-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Body, Fuze	3 OF 3	9333824

NEXT HIGHER ASSEMBLY: 9347183/9347185

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor (cont.)				
217	Thickness of flange	Level IV	3.4	Gage
218	Chamfer on flange	Level IV	3.4	Gage
219	Diameter of safety pin hole	Level IV	3.4	Gage
220	Orientation of notches (First Article only)	Level IV	3.4	Gage
221	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONF MANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Spring, slider	1 OF 1	9333811

NEXT HIGHER ASSEMBLY: 9333825/9345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Spring load requirement, max.	Level III	3.4	Gage
102	Spring load requirement, min.	Level III	3.4	Gage
103	Solid height	Level III	3.4	Gage
104	Outside diameter of spring	Level III	3.4	Gage
105	Spring embrittlement	Level III	3.4.4.1	4.5.2
106	Spring set	Level III	3.4.4.2	4.5.3
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Screw, Pivot	1 OF 1	9333808

NEXT HIGHER ASSEMBLY: 9333825/9345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Distance from pivot hole bottom to underside of screw head	Level III	3.4	Gage
102	Diameter of pivot hole	Level III	3.4	Gage
Minor				
201	Length of thread to underside of head	Level IV	3.4	Gage
202	Diameter of tapered counterbore at bottom	Level IV	3.4	Gage
203	Diameter of tapered counterbore at top	Level IV	3.4	Gage
204	Evidence of Poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Anti-twist Pin	1 OF 1	9347220

NEXT HIGHER ASSEMBLY: 9347184/9347185

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Condition	Level III	3.4	Cert.
102	Pin diameter	Level III	3.4	Gage
103	Length of Pin	Level III	3.4	Gage
104	Surface finish	Level III	3.4	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Sleeve, Fuze Body	1 OF 1	9345131

NEXT HIGHER ASSEMBLY: 9347185

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Inside diameter	Level III	3.4	Gage
102	Shear wire hole location	Level III	3.4	Gage
103	Shear wire hole diameter	Level III	3.4	Gage
Minor				
201	Body diameter	Level IV	3.4	Gage
202	True position, I.D. to O.D.	Level IV	3.4	Gage
203	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Fuze Body Assembly	1 OF 1	9347185

NEXT HIGHER ASSEMBLY: 9333825/9345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Position of staked sleeve	Level III	3.4	Gage
102	Pins above bottom of slider slot	Level III	3.4	Gage
103	Anti-twist pin pushout requirement	Level III	3.4	Gage
104	Adhesive in not called for area	Level III	3.4	Visual
105	Wrong adhesive	(a)	3.4	Certification
Minor				
201	Sleeve pushout requirement	Level IV	3.4	Gage
202	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:
(a) Verify prior to use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Trigger	1 OF 1	9333812

NEXT HIGHER ASSEMBLY: 9333825/9345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Spherical radius	Level III	3.4	Gage
102	Overall length	Level III	3.4	Gage
103	Small tip diameter	Level III	3.4	Gage
104	Major body diameter	Level III	3.4	Gage
105	Length of pivoting shank	Level III	3.4	Gage
106	True position of true diameter to major body diameter	Level III	3.4	Gage
107	Finish on tip	Level S-4	3.4	Gage
Minor				
201	Length of major O.D.	Level IV	3.4	Gage
202	Relieved diameter	Level IV	3.4	Gage
203	True position of pivoting diameter to major body diameter	Level IV	3.4	Gage
204	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Pin, Arming	1 OF 1	9333814

NEXT HIGHER ASSEMBLY: 9333825/9345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Flange diameter	Level III	3.4	Gage
102	True position of blind holes	Level III	3.4	Gage
103	Depth of blind holes	Level III	3.4	Gage
104	Shank diameter	Level III	3.4	Gage
105	Tip length	Level III	3.4	Gage
106	Shear wire hole diameter	Level III	3.4	Gage
107	Thread improper	Level III	3.4	Gage
108	Location of shear wire hole from flange	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Wire, Shear	1 OF 1	9345128

NEXT HIGHER ASSEMBLY: 9333825/9345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Shear load	Level III	3.4	Gage
Minor				
201	Wire length	Level IV	3.4	Gage
202	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFC IANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Slider, M55 Detonator Assembly	1 OF 1	9347219

NEXT HIGHER ASSEMBLY: 9333825/9345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Orientation of detonator	Level III	3.4	Visual
102	Excessive adhesive	Level III	3.4	Visual
103	Detonator deterioration	Level III	3.4.5	4.5.4
104	Wrong adhesive	(a)	3.4	Certification
105	Adhesive in not-called-for area	Level III	3.4	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:
(a) Verify prior to use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH 4.4.2.12	TITLE Pin, Safety	SHEET 1 OF 1	DRAWING NUMBER 9333818
			NEXT HIGHER ASSEMBLY 9333825/9345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Wire diameter	Level III	3.4	Gage
102	Length end to center of locking loop	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Pinion, Escapement	1 OF 1	8833366
			NEXT HIGHER ASSEMBLY: 8833364

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Pinion profile	Level III	3.4	Gage
102	Plating improper	Level III	3.4	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH 4.4.2.14	TITLE Gear, Escapement	SHEET 1 OF 1	DRAWING NUMBER 8833365
			NEXT HIGHER ASSEMBLY 8833364

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Concentricity	Level III	3.4	Gage
102	Gear profile	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFC ANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Escapement Gear-Pinion Assembly	1 OF 1	8833364
			NEXT HIGHER ASSEMBLY: 8833353

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Gap between gear and pinion	Level III	3.4	Gage
102	Staking improper	Level III	3.4	Visual/Gage
103	Perpendicularity	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Pinion, No. 1	1 OF 1	8833360

NEXT HIGHER ASSEMBLY: 8833358

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Pinion profile	Level III	3.4	Gage
102	Plating improper	Level III	3.4	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.17	Gear, No. 1	1 OF 1	8833359

NEXT HIGHER ASSEMBLY: 8833358

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Gear profile	Level III	3.4	Gage
102	Plating improper	Level III	3.4	Visual/Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	No. 1 Gear-Pinion Assembly	1 OF 1	8833358

NEXT HIGHER ASSEMBLY: 8833353

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Staking improper	Level III	3.4	Gage/Visual
102	Perpendicularity	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.19	Pinion No. 2	1 OF 1	8833363

NEXT HIGHER ASSEMBLY: 8833361

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Pinion profile	Level III	3.4	Gage
102	Plating improper	Level III	3.4	Gage/Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.20	Gear, No. 2	1 OF 1	8833362

NEXT HIGHER ASSEMBLY: 8833361

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Gear profile	Level III	3.4	Gage
102	Plating improper	Level III	3.4	Gage/Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	No. 2 Gear-Pinion Assembly	1 OF 1	8833361
			NEXT HIGHER ASSEMBLY 8833353

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Gap between gear and pinion	Level III	3.4	Gage
102	Staking improper	Level III	3.4	Gage/Visual
103	Perpendicularity	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.22	Lever	1 OF 1	8833369

NEXT HIGHER ASSEMBLY: 8833367

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Diameter	Level III	3.4	Gage
102	Lobe width	Level III	3.4	Gage
103	Pallet hole position from center	Level III	3.4	Gage
104	Pallet hole separation	Level III	3.4	Gage
105	Pallet hole diameter	Level III	3.4	Gage
106	Center hole diameter	Level III	3.4	Gage
Minor				
201	Lever thickness	Level IV	3.4	Gage
202	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.23	Pallet, Lever	1 OF 1	8833368

NEXT HIGHER ASSEMBLY: 8833367

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Pallet diameter	Level III	3.4	Gage
102	Plating improper	Level III	3.4	Gage/Visual
Minor				
201	Pallet length	Level IV	3.4	Gage
202	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Shaft, Lever	1 OF 1	8833370

NEXT HIGHER ASSEMBLY: 8833367

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Surface finish improper	Level III	3.4	Gage
102	Plating improper	Level III	3.4	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.25	Lever Assembly	1 OF 1	8833367

NEXT HIGHER ASSEMBLY: 8833353

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Staking improper	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-02-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.26	Plate, Bottom	1 OF 1	8833356

NEXT HIGHER ASSEMBLY: 8833353

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Location of pinion holes	Level III	3.4	Gage
102	Diameter of pinion holes	Level III	3.4	Gage
103	Location of pillar holes	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONI MANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.27	Plate, Top	1 OF 1	8833357

NEXT HIGHER ASSEMBLY: 8833353

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Location of pinion holes	Level III	3.4	Gage
102	Location of tabs from front flat edge	Level III	3.4	Gage
103	Diameter of pinion holes	Level III	3.4	Gage
104	Location of pillar holes	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.28	Pillar	1 OF 1	8833355

NEXT HIGHER ASSEMBLY: 8833353

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Major diameter length	Level III	3.4	Gage
102	Minor diameter	Level III	3.4	Gage
103	Concentricity of minor diameter	Level III	3.4	Gage
Minor				
201	Major diameter	Level IV	3.4	Gage
202	Length of minor diameter	Level IV	3.4	Gage
203	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.29	Escapement Gear Assembly	1 OF 1	8833353

NEXT HIGHER ASSEMBLY: 9333825

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Axial pull of assembly	Level S-4	3.4	Gage
102	Maximum height	Level III	3.4	Visual
103	Wrong lubricant	(a)	3.4	Certification
104	Lubricant missing	Level III	3.4	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:
(a) Verify prior to use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.30	Body, Firing Pin	1 OF 1	9333816

NEXT HIGHER ASSEMBLY: 9345130

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Large diameter	Level III	3.4	Gage
102	Detent ball angle	Level III	3.4	Gage
104	Dowel pin hole diameter	Level III	3.4	Gage
Minor				
202	Concentricity, stem to large diameter	Level IV	3.4	Gage
204	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.31	Pin, Straight	1 OF 1	9345321

NEXT HIGHER ASSEMBLY: 9345130

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Tip diameter	Level III	3.4	Gage
102	Tip edge radius, max.	Level III	3.4	Gage
103	Tip length	Level III	3.4	Gage
104	Tip angle	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.32	Firing Pin, Subassemby	1 OF 1	9345130

NEXT HIGHER ASSEMBLY: 9347182

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major	None defined.			
Minor				
201	Flush to above	Level IV	3.4	Gage
202	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.33	Cap, Firing Pin	1 OF 1	9333817

NEXT HIGHER ASSEMBLY: 9347182

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Head diameter	Level III	3.4	Gage
102	Inside diameter	Level III	3.4	Gage
103	True position between head diameter and inside diameter	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.34	Spring, Firing Pin	1 OF 1	9333810

NEXT HIGHER ASSEMBLY: 9347182

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Spring load requirement, max.	Level III	3.4	Gage
102	Spring load requirement, min.	Level III	3.4	Gage
103	Solid height	Level III	3.4	Gage
104	Outside diameter of spring	Level III	3.4	Gage
105	Spring embrittlement	Level III	3.4.4.1	4.5.2
106	Spring set	Level III	3.4.4.2	4.5.3
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.35	Firing Pin Assembly	1 OF 1	9347182

NEXT HIGHER ASSEMBLY: 9333825/9345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Integrity of firing pin point	Level III	3.4	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

AMSMC FORM 1570b-E, 1 Jul 89

Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH 4.4.2.36	TITLE Ball, Detent	SHEET 1 OF 1	DRAWING NUMBER 9349223
			NEXT HIGHER ASSEMBLY 9333825/9345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Ball diameter	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFO ANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.37	Plate, Barrier	1 OF 1	9333809

NEXT HIGHER ASSEMBLY: 9345168/9345166

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Outside diameter	Level S-4	3.4	Gage
102	Thickness	Level III	3.4	Gage
103	Depth of lead hole	Level III	3.4	Gage
104	Flatness	Level S-4	3.4	Gage
Minor				
201	True position, Lead hole to O.D.	Level IV	3.4	Gage
202	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.38	Spring, Arming Lock Pin	1 OF 1	9345129

NEXT HIGHER ASSEMBLY: 9333825/9345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Wire diameter	Level III	3.4	Gage
Minor				
201	Width of opening	Level IV	3.4	Gage
202	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.39	Disc, Closing	1 OF 1	9333821

NEXT HIGHER ASSEMBLY: 9333823

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Disc thickness	Level III	3.4	Gage
Minor				
201	Diameter	Level IV	3.4	Gage
202	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.40	Lead Cup, Short	1 OF 1	9333822

NEXT HIGHER ASSEMBLY: 9333823

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Overall length	Level III	3.4	Gage
102	Material thickness	Level III	3.4	Gage
103	Major diameter of closing disk hole	Level III	3.4	Gage
104	Concentricity between I.D. of cup and closing disc hole	Level III	3.4	Gage
Minor				
201	Overall flange diameter	Level IV	3.4	Gage
202	Inside diameter of cup	Level IV	3.4	Gage
203	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.41	Lead Cup Assembly, Short	1 OF 1	9333823

NEXT HIGHER ASSEMBLY: 9345166

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Explosive height	Level III	3.4	Gage
102	Overall height	Level III	3.4	Gage
103	Lead output	32	3.4.6	4.5.5
Minor				
201	Outside diameter of flange	Level IV	3.4	Gage
202	Outside diameter	Level IV	3.4	Gage
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.42	Lead Cup, Long	1 OF 1	9333819

NEXT HIGHER ASSEMBLY: 9333820

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Overall length	Level III	3.4	Gage
102	Material thickness	Level III	3.4	Gage
Minor				
201	Outside diameter of flange	Level IV	3.4	Gage
202	Inside diameter of cup	Level IV	3.4	Gage
203	Outside diameter	Level IV	3.4	Gage
204	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFO IANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH 4.4.2.43	TITLE Lead Cup Assembly, Long	SHEET 1 OF 1	DRAWING NUMBER 9333820
			NEXT HIGHER ASSEMBLY 9345166

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Overall height	Level III	3.4	Gage
102	Explosive height	Level III	3.4	Gage
103	Lead output	32	3.4.6	4.5.5
Minor				
201	Outside diameter of flange	Level IV	3.4	Gage
202	Outside diameter	Level IV	3.4	Gage
203	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.44	Plate Assembly, Barrier (M230)	1 OF 1	9345166

NEXT HIGHER ASSEMBLY: 9333825

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Position of short lead	100%	3.4	Visual
102	Damaged lead cup	100%	3.4	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFO .ANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.45	Screw, Modified Drive	1 OF 1	9390620

NEXT HIGHER ASSEMBLY: 9333825/345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Thread body diameter	Level III	3.4	Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH 4.4.2.46	TITLE Barrier Plate Assembly with Retainer Tape	SHEET 1 OF 1	DRAWING NUMBER 12561493
			NEXT HIGHER ASSEMBLY 9333825/9345168

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Tape properly secured	100%	3.4	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFC ANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH 4.4.2.47	TITLE Fuze General Purpose Grenade, Practice: M231	SHEET 1 OF 2	DRAWING NUMBER 9345168
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Special (a)				
1	Safety pin missing	100%	3.4	AAIE (g)
2	Presence of shear wire	100%	3.4	AAIE (g)
3	Safety pin pull test	64 (c)	3.5.8	4.5.13
4	Shear wire pull test	64 (c)	3.5.7	4.5.12
5	Arming pin/slider lock	100%	3.5.12	4.5.17
6	Auto trigger	100%	3.5.13	4.5.17
7	Fuzing No-Go	32 (c)	3.5.10	4.5.15(d)
8	Transportation vibration	12	3.5.3	4.5.8/4.5.15(b)/4.5.16(b)
9	Helicopter vibration	12	3.5.4	4.5.9/4.5.15(b)/4.5.16(b)
10	Five foot drop	15	3.5.5	4.5.10
11	Thermal shock	12	3.5.6	4.5.11/4.5.15(b)/4.5.16(b)

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

ARAGRAPH 4.4.2.47	TITLE Fuze General Purpose Grenade, Practice: M231	SHEET 2 OF 2	DRAWING NUMBER 9345168
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
101	Major diameter of fuze	Level III	3.4	Gage
102	Height of major diameter of fuze	Level III	3.4	Gage
103	Arming time	64 (c)	3.5.9	4.5.14(e)
104	Fuzing Go	32 (c)	3.5.11	4.5.16(d)
105	Torque requirement	115	3.4	Gage
106	Wrong adhesive	(f)	3.4	Certification
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:
(a) See MIL-A-48078 for definition. (b) Fuzing No-Go -1/2 test sample at -50° F ± 5° F, 1/2 test sample at 160° F ± 5° F; Fuzing Go - 1/2 test sample at -50° F ± 5° F, 1/2 test sample at 160° F ± 5° F. (c) The same fuzes are to be subjected to these tests. If the test within 90 seconds requirement described below cannot be maintained, however, each test or any combination of these tests performed on a separate group of fuzes which could increase the maximum sample size. (d) Prior to performing the test, condition fuzes for two hours after stabilization at indicated temperatures: -50° F ± 5° F -16 fuzes; +150° F ± 5° F, 16 fuzes. Tests shall be completed within 90 seconds of removal from temperature conditioning chambers. All fuzes tested at cold shall be conditioned while sealed in plastic bags. (e) Prior to performing the test, condition fuzes for two hours after stabilization at indicated temperatures: -50° F ± 5° F -32 fuzes; +150° F ± 5° F, 32 fuzes. Tests shall be completed within 90 seconds of removal from temperature conditioning chambers. All fuzes tested at cold shall be conditioned while sealed in plastic bags. (f) Verify prior to use. (g) Automated Acceptance Inspection Equipment in accordance with MIL-A-70625.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.48	TITLE Fuze, General Purpose Grenade: M230	SHEET 1 OF 2	DRAWING NUMBER 9333825

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
Critical	None defined.			
Special (a)				
1	Safety pin missing	100%	3.4	AAIE (g)
2	Presence of shear wire	100%	3.4	AAIE (g)
3	Safety pin pull test	64 (c)	3.5.8	4.5.13
4	Shear wire pull test	64 (c)	3.5.7	4.5.12
5	Arming pin/slider lock	100%	3.5.12	4.5.17
6	Auto trigger	100%	3.5.13	4.5.17
7	Fuzing No-Go	32 (c)	3.5.10	4.5.15(d)
8	Transportation vibration	12	3.5.3	4.5.8/4.5.15(b)/4.5.16(b)
9	Helicopter vibration	12	3.5.4	4.5.9/4.5.15(b)/4.5.16(b)
10	Five foot drop	15	3.5.5	4.5.10
11	Thermal shock	12	3.5.6	4.5.11/4.5.15(b)/4.5.16(b)

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-F-63446B (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.48	Fuze, General Purpose Grenade: M230	2 OF 2	9333825

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	NEXT HIGHER ASSEMBLY / INSPECTION METHOD REFERENCE
Major				
101	Major diameter of fuze	Level III	3.4	Gage
102	Height of major diameter of fuze	Level III	3.4	Gage
103	Arming time	64 (c)	3.5.9	4.5.14(e)
104	Fuzing Go	32 (c)	3.5.11	4.5.16(d)
105	Torque requirement	115	3.4	Gage
106	Wrong adhesive	(f)	3.4	Certification
107	Damaged lead	100%	3.4	Visual
108	Presence of lead retainer	100%	3.4	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.7	Visual

NOTES:
(a) See MIL-A-48078 for definition. (b) Fuzing No-Go -1/2 test sample at -50° F ± 5° F, 1/2 test sample at 160° F ± 5° F; Fuzing Go - 1/2 test sample at -50° F ± 5° F, 1/2 test sample at 160° F ± 5° F. (c) The same fuzes are to be subjected to these tests. If the test within 90 seconds requirement described below cannot be maintained, however, each test or any combination of these tests performed on a separate group of fuzes which could increase the maximum sample size. (d) Prior to performing the test, condition fuzes for two hours after stabilization at indicated temperatures: -50° F ± 5° F -16 fuzes; +150° F ± 5° F, 16 fuzes. Tests shall be completed within 90 seconds of removal from temperature conditioning chambers. All fuzes tested at cold shall be conditioned while sealed in plastic bags. (e) Prior to performing the test, condition fuzes for two hours after stabilization at indicated temperatures: -50° F ± 5° F -32 fuzes; +150° F ± 5° F, 32 fuzes. Tests shall be completed within 90 seconds of removal from temperature conditioning chambers. All fuzes tested at cold shall be conditioned while sealed in plastic bags. (f) Verify prior to use. (g) Automated Acceptance Inspection Equipment in accordance with MIL-A-70625.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.4.3 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is identified in "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and section 6.4 herein.

4.5 Methods of inspection. PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rides, regulations and procedures must be followed in handling and processing of these materials.

4.5.1 Porosity check A random sample shall be chosen from each cavity of an incoming lot of Fuze Body die cast blanks in accordance with 4.4.2c. The castings shall then be machined to final configuration and inspected for size, location, depth and frequency of surface defects. The acceptable limit of defects is described in both Table III and Figure 1. If the number of rejects from a given cavity exceeds the level reported in 4.4.2c, the incoming quantity from that cavity may either be rejected and removed from production or screened 100% after machining. A cavity, once pulled from production, cannot be used until requalified. Sectioning maybe performed on any portion of the inspection sample if necessary. The firing pin bore shall be inspected 100% by using an Otoscope to direct light and magnify the image.

4.5.2 Spring embrittlement. The spring shall be stretched along its longitudinal axis until the spring no longer returns to its original position. When the spring no longer returns to its original length or profile, it shall be considered distorted permanently. Any spring that breaks prior to this point shall be classed defective. All samples subjected to this test will be damaged and shall not be returned to the production lot.

4.5.3 Spring set. The spring shall be assembled in a restraining fixture in accordance with 4.4.3 and compressed to a nominal height at .430 inch, placed in an oven and subjected to a temperature of not more than 175° F ± 5° F for 24 hours. The springs shall be removed from the oven and be cooled to the ambient temperature. After cooling the springs for 2 hours minimum, the springs shall be measured to ascertain compliance with the specified drawing load requirements. Any spring that fails the requirement shall be classed defective.

4.5.4 Detonator deterioration. If the total time between original acceptance of any detonator lot and the assembly of that lot and sliders exceeds two years, or if the detonators have been subjected to adverse conditions, however brief, at any time since the previous tests, the detonator lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-D-14978. This test shall be performed by the contractor on detonators selected by the government inspector.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

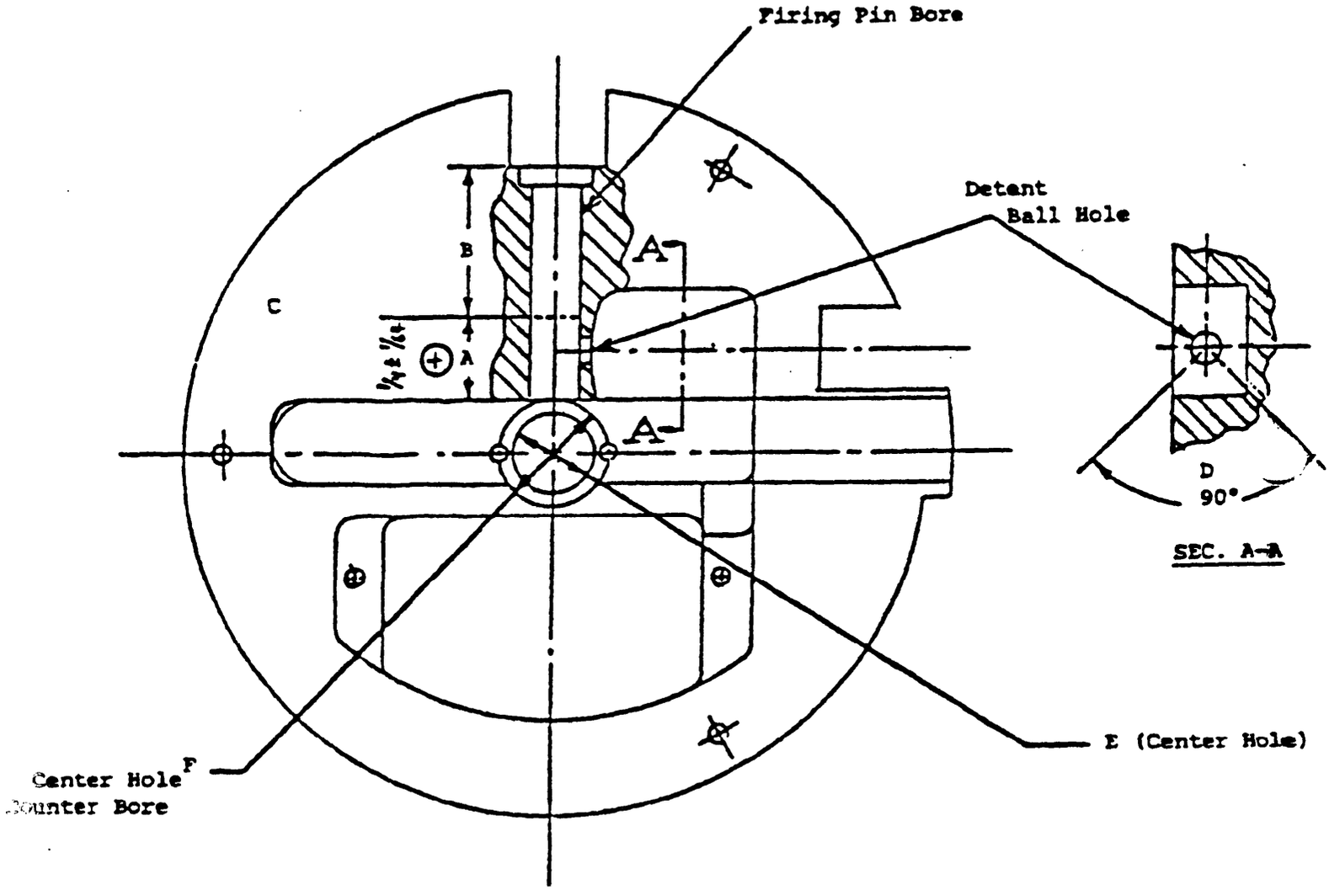


Figure 1. Defect zones.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

TABLE III. Maximum allowable pore size and frequency per location.

ZONE A		ZONE B	
SIZE	FREQUENCY	SIZE	FREQUENCY
.040-.060	1	.040-.060	1
.010-.040	see note 1	.010-.040	see note 1
.000-.010	no limit	.000-.010	no limit

ZONE C		ZONE D	
SIZE	FREQUENCY	SIZE	FREQUENCY
.010-.040	see note 1		
.000-.010	no limit	.000-.010	4, see note 2

ZONE E		ZONE F	
SIZE	FREQUENCY	SIZE	FREQUENCY
.040-.060	2		
.010-.040	see note 1		
.000-.010	no limit	.000-.010	4

NOTE 1: Surface defects having a maximum linear dimension of up to .040 are allowed throughout the casting (ZONE A, B, C, D and E) except in the detent ball hole (ZONE D) and the center hole counter bore (ZONE F). A sum total of 50 is allowed through ZONES A, B, C, and E.

NOTE 2: Permissible defects in the detent ball hole must not extend into the edges of the hole.

DEFINITIONS:

ZONE A - The length of the firing pin bore (.158''+.003" in diameter) extending back from the slider slot a distance of 1.4" + 1/64" (see figure 1).

ZONE B - The length of the firing pin bore extending back from the end of ZONE A to and including all surfaces of the firing pin cap counterbore (.218"+.003" diameter). (see figure 1).

ZONE C - All other surfaces of the casting excluding the detent ball hole (ZONE D) and the center hole counterbore (ZONE F).

ZONE D - The surface of the detent ball hole included within the arc of 90°, the center of which is the nominal contact point of the detent ball (see figure 1).

ZONE E - The surface of center hole (.237''+.003" diameter).

ZONE F - The surface of center counterbore (.313"+.004" diameter).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

4.5.5 Lead output. A random sample of Short (P/N 9333823) and Long (P/N 9333820) Lead Cup Assemblies shall be selected. Each Long and Short Cup Assembly shall be assembled to a holding fixture, to be approved by the government, that simulates relative lead position in a fuze. Design of the future shall be such as to enable the output end of the Long Lead Cup Assembly to locate within .010 inch of a steel witness plate. The fixture shall also house an M55 detonator positioned .04 ± .01 inch above the input end of the Short Lead Cup Assembly. Initiation of the detonator shall result in high order propagation of the Short and Long Lead Assemblies. After the test, the witness plate shall be inspected, using appropriate measuring equipment, for depth of impression. The depth shall be a minimum of .015 inch. Any output that results in an impression of less than the requirement shall be considered a failure. A single rejection shall fail the lot.

4.5.6 Jolt. A sample of fuzes shall be randomly selected from the first article sample and be prepared for the jolt test. The fuzes shall be mounted to special holding fixtures that simulate the fuze seat in a grenade body. The holders shall be designed such as to allow for assembly of the felt take-up pad beneath the fuze assembly compressed to the proper degree. Clamping arrangements, rather than stakes, shall suffice for securing the fuze assembly to the holder providing the fuze is properly seated and remains so for the duration of the test. Fuzes shall be tested without the RAD Assembly Complete (P/N 9345154) attached to the arming pin. The safety pin (P/N 93338 18) shall be in place. After the test, the fuzes shall be inspected prior to the removal from the holding fixture and it shall be established that the fuzes are safe to store, handle, use or dispose of. In order to be considered safe to use, the fuze assembly must meet one of the following requirements:

a. The safety pin must be engaged and the fuze not armed. The shear wire may be cut or the firing pin triggered and the fuze shall still be considered safe to handle, store, dispose of or use.

b. The shear wire must not be cut and the fuze not armed. The safety pin maybe disengaged or the firing pin triggered and the fuze shall be safe to handle, store, dispose of or use.

If any fuze fails to meet either requirement it shall be considered a safety failure. A single safety failure shall reject the lot. Any fuze subjected to this test will be damaged and shall not be returned to the production lot.

4.5.7 Jumble. A sample of fuzes shall be randomly selected from the first article sample and subjected to the jumble test. The fuzes shall be tested in a fixture simulating confinement in the M261/M267 Warhead. The fuze with RAD assembly complete shall be mounted to an inert loaded M73/M75 Grenade body. The future shall facilitate the testing of the grenade in the number 1 position as currently defined in the warhead. The tested grenade shall be sandwiched between the grenade above and the fuze support below. After the test, it shall be established that the fuze is safe to handle, store, use or dispose of. In order to be considered safe, the fuze must meet all of the following requirements:

a. The shear wire must not be broken.
b. The slider must not be broken.
c. The explosive element must not have been initiated or broken.

The first article shall be rejected if one or more fuzes fail to meet any of the above requirements or are consideredl unsafe to handle, store, use or dispose of.

4.5.8 Transportation vibration. A sample of fuzes shall be randomly selected from each production lot and subjected to the Transportation Vibration Test. Half of the fuzes shall be tested at -65°F and the remaining half at 160° F. Temperature preconditioning shall be four

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

hours minimum. In the preparation for the test, each fuze shall be secured to holding fixtures that simulate the fuze seat of a grenade body. Design of the fixtures shall be such as to allow for assembly of a felt take-up pad beneath the barrier plate of the fuze compressed to the proper degree. Clamping arrangements shall suffice as a method to join the fuze to the holding fixture provided the fuze is properly seated and remains so for the duration of the test. It shall be established that the clamping apparatus does not alter the resonant frequencies of the fuze assembly. Prior to the test the safety pin shall be removed from the fuzes. After the test, prior to removing the fuzes from the holding fixtures, it shall be established visually that the fuzes are safe. To be considered safe the fuze must meet each of the following requirements:

a. The shear wire must not be cut.
b. The slider must be in the full safe position.
c. The firing pin must not have triggered.

If any fuze fails to meet any requirement listed above, it shall be considered a safety failure. A single safety failure shall reject the lot.

Having established the fuzes to be safe, operability shall be verified. Half of the samples tested at hot and half of those tested at cold temperature shall be subjected to the Fuze No-Go test. Any fuze that fails the Fuzing No-Go test shall be considered an operability failure. A single such failure shall reject the lot. Fuzes that pass the Fuzing No-Go test shall be disposed of per 4.5.15. The remaining fuzes tested at hot and the remaining fuzes tested at cold shall be subjected to the Fuzing Go test. Fuzes that fail to function shall be considered operability failures. See 4.5.16 for the quantity of Fuzing Go failures that will reject the lot.

4.5.9 Helicopter vibration. A sample of fuzes shall be randomly selected from each production lot and subjected to&e sinusoidal cycling described by Curve M in Method 514 of MIL-STD-810B. All fuzes shall be tested at ambient temperature. Prior to the test, the fuzes shall be secured to holding fixtures that simulate the fuze seat of a grenade body assembly. Design of the fixtures shall be such as to allow for assembly of a felt take-up pad compressed to the proper degree. The methods utilized to attach the fuze to the holding fixture shall not alter the resonant frequencies of the fuze. The safety pin shall be removed from the fuze just before the test is begun. Operability verification and acceptance criteria shall be identical to those requirements specified in 4.5.8 (Transportation Vibration).

4.5.10 Five foot drop. A random sample of fuzes shall be selected for the drop test as specified in MIL-STD-331. The drops shall be made in the following orientations:

a. Base down
b. 45° Base down
c. Horizontal
d. 45° Nose down
e. Nose down

One fifth of the fuzes shall be dropped in each orientation and thereafter it shall be established visually that the fuze continues to be safe to handle and dispose of. For the fuze to be considered safe, the following conditions must be met:

a. The safety pin must be engaged and shown to have locked the slider in the full safe position. If the safety pin remains locked in its original position (i.e., the locking loop of the pin engaged with the safety pin post protruding from the fuze body) and it has been established that the detonator is out of line with the firing pin, then the fuze shall be considered safe to handle and dispose of even if the following is evident:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:27Z
Check the source to verify that this is the current version before use.

(1) The shear wire is cut
(2) The firing pin has triggered

b. If the safety pin has removed itself from the fuze during the drop test, the fuze shall be considered safe to handle and dispose of only if all of the following conditions are met:

(1) The shear wire has not been cut.
(2) The slider has not moved from the full safe position (i.e., the bore rider remains completely contained within the fuze body).
(3) The firing pin has not triggered (i.e., the tail of the firing pin continues to protrude outside the fuze beyond the firing pin cap).

Should any fuze tested fail to meet condition a. or b, it shall be considered a safety fuze failure. A single safety failure shall reject the lot. Any fuze subjected to this test will be damaged and shall not be returned to the production lot.

4.5.11 Thermal shock. A random sample of fuzes shall be selected from each production lot and subjected to the Thermal Shock Test C7of MIL-STD-331. Each fuze shall be sealed in a plastic bag prior to the test. After completion of the test, it shall be visually determined that neither the shear wire nor the pin has broken or disengaged. Any such failure shall be considered a safety failure and shall reject the lot. Operability shall be established by subjecting half the fuzes to the Fuzing No-Go test and the remaining half to the Fuzing Go test. Any fuze that fails to pass shall be considered an operability failure. A single Fuzing No-Go failure shall reject the lot. See 4.5.16 for the quantity of Fuzing Go failures that will reject the lot.

4.5.12 Shear wire pull test. A random sample of fuzes shall be selected from each production lot and subjected to the Shear Wire Pull test. The fuzes shall be mated to a holding fixture that allows an axial load to be applied to the arming pin. A load ceil shall be utilized to measure the applied force within 0.1 pound. The load applying device must cause axial displacement of the arming pin at a constant rate of .040 ± .030 inches per second. Any fuze of the sample lot that requires less than 16 pounds or more than 22 pounds to cut the shear wire shall be considered a failure. One failure below 15.5 pounds shall cause rejection of the lot. One failure above 22.0 pounds shall trigger a retest. One failure during retest shall cause rejection of the lot.

4.5.13 Safety pin pull test. A random sample of fuzes shall be selected from each production lot and subjected to the Safety Wire Pull test. Prior to the test, the fuze shall be assembled to a nest that will secure it traversely. It shall also be established visually that the shear wire is intact before the pull testis performed. A load shall be applied to the safety pin and the force required to remove it from the fuze measured. The load applying device must cause the axial retraction of the safety pin from a constant rate of .040 ± .030 inches per second. An effort less than 1 (one) pound or greater than 8 pounds shall be considered a failure. One failure below one pound or two failures above 8 pounds shall reject the lot.

4.5.14 Arming time. A random sample of fuzes shall be taken from each production lot Each fuze shall be assembled to a fixture that enables determination of the time required for the slider to complete its arming stroke, which commences at the arming pin retraction and ends with uncaging of the trigger. The design of the fixture shall be government approved. Upon release, the slider must move into the anneal position in 1.50 ± .50 seconds. A fuze that fails to arm within the prescribed time interval shall be considered a reject. A single reject below minimum arming time shall reject the production lot. Two rejects above the maximum time or two jam-ups shall reject the production lot. One jam-up shall result in a retest of 64 fuzes (32 hot, 32 cold). One jam-up during the retest shall reject the production lot. Two rejects above the maximum

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

time or one jam-up shall reject a First Article sample ANY TIME IN EXCESS OF 3 SECONDS SHALL BE CONSIDERED A JAM-UP.

4.5.15 Fuzing No-Go Test A random sample of fuzes shall be selected from each production lot and subjected to and pass Fuzing No-Go Test.

WARNING: This test is dangerous and all necessary safety precautions shall be taken. Test operators shall be adequately shielded from the fuzes and their holding fixtures during arming, application of impulse and re-safing.

Remove shear wire, if not already removed due to any previous test, prior to mounting the fuze to the holding fixture. After mounting the fuze to the holding fixture, but prior to performing the test the safety pin shall be removed. Removal of the safety pin shall be considered as the beginning of the arming operation and thus safety shielding requirement applies. The fuze shall be secured in a holding fixture in such a manner that the shock impulse that the fuze senses will not be attenuated or magnified beyond that applied. The design of the fixture shall be such as to not support the fuze beneath the barrier plate except in the area of the fuze seat. Arming of the fuze is accomplished by raising of the arming pin, removal of the safety pin, and allowing the slider to move to its fully armed position. With the fuze in the armed state, an impulse shall be applied to the fuze/fixture in a direction parallel to the axis of the fuze within one degree. The magnitude of the impulse shall be 180 ± 20 gravities. Duration of the impulse shall be 1.5 ±.5 milliseconds. The orientation of the fuze with respect to the impulse shall be such as to simulate a normal target impact. The trigger shall not release the detent ball and firing pin as a result of the shock impulse. Any fuze that functions during the No-Go test shall be classified as a critical failure and disposed of in accordance with proper safety procedures. Fuzes that pass the No-Go test shall be resafed by moving the slider back into its fully safe position and lowering the arming pin until it is fully seated. The safety pin is reinserted after the slider is back in to its fully safe position. Resafing is considered complete at the end of installation of the safety pin. Any fuze that cannot be properly resafed shall be disposed of in accordance with proper safety procedures. Resafed No-Go fuzes at the end of the Go test will be forced to function after remounting in holding fixture or will be disposed of in accordance with proper safety procedures.

4.5.16 Fuzing Go Test. A random sample of fuzes shall be selected from each production lot and subjected to the Fuzing Go Test.

WARNING: This test is dangerous and all necessary safety precautions shall be taken. Test operators shall be adequately shielded from the fuzes and their holding fixtures during arming, application of impulse and resafing.

The fuze shall be secured in a holding fixture in such a manner that the shock impulse that the fuze senses will not be attenuated or magnified beyond that applied. The design of the fuze fixture shall be as such as to not support the fuze beneath the barrier plate except in the area of the fuze seat. Remove shear wire prior to mounting the fuze in the holding fixture. After mounting the fuze in the holding fixture, but prior performing the test, the arming pin shall be raised and the safety pin shall be removed. Removal of the safety pin shall be considered as beginning of the arming operation and thus the shielding operation applies. With the fuze in the armed state, an impulse shall be applied to the fuze/fixture in a direction parallel to the axis of the fuze within one degree. The magnitude of the impulse shall be 700 gravities maximum. Duration of the impulse shall be 1.0 ±.5 milliseconds. The orientation of the fuze with respect to the impulse shall be such as to simulate a normal target impact The trigger shall release the detent ball and firing pin causing initiation of the explosive train. Any fuze that fails to function shall be considered a failure. A sum total (for all Fuzing Go Test) of three failures shall reject the production lot. Two failures shall result in a retest (including transportation vibration, thermal shock, helicopter vibration, and temperature conditioning) of 50 fuzes. One failure during the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

retest shall reject the production lot. Two failures shall reject the first article sample. Any fuze that fails the Fuzing Go Test shall be forced to function or disposed of in accordance with proper safety procedures before removal from the fixture. This can be accomplished by increasing the magnitude of the impulse to approximately 1,000 gravities. Should the armed fuze still not trigger, a hang fire has occurred and must be handled accordingly. Revision must be made that will allow for removal of the failed fuze from the test equipment while maintaining a shielded condition.

4.5.17 Arming pin/slider lock and auto trigger test. Each fuze shall be subjected to and pass the Arming Pin/Slider Lock and Auto Trigger test.

WARNING: This test is dangerous and all necessary safety precautions shall be taken. Test operators shall be adequately shielded from the fuzes and their holding fixtures during arming and resafing.

Remove shear wire if necessary prior to mounting the fuze in the holding fixture. After mounting the fuze to the holding fixture, but prior to performing the test, the safety pin shall be removed. Removal of the safety pin shall be considered as beginning of the arming operation and thus safety shielding requirement applies. Upon removal of the safety pin, observe that the slider remains in the fully safe position. Should the fuze arm upon removal of the safety pin, it shall be rejected as a critical failure and disposed of in accordance with proper safety procedures. If the slider remains in the safe position upon removal of the safety pin, the fuze shall then be allowed to arm by raising the arming pin. The trigger shall not spontaneously release the detent ball and firing pin when the slider moves to the fully armed position. Any fuze that fails this requirement shall be classified as a critical failure and disposed of in accordance with proper safety procedures. Fuzes that pass the Auto Trigger test shall be resafed by moving the slider back into its fully safe position and lowering the arming pin until it is fully seated. The safety pin is reinserted after the slider is back in to its fully safe position. Resafing is considered complete at the end of installation of the safety pin. Any fuze that cannot be properly resafed shall be disposed of in accordance with proper safety procedures.

4.5.18 Data requirements. The contractor shall submit a quality inspection test demonstration and evaluation report in accordance with 6.3.

5. PACKAGING

5.1 Special packaging requirement. To protect the escapement gear assembly from part particulate contamination, each fuze shall be sealed in a plastic bag.

5.2 **Packaging**. Fuzes shall be packaged in accordance with Dwg. 9340491.

5.3 Packing. Fuzes shall be packed in accordance with Dwg. 9340492.

5.4 **Marking**. Marking shall be in accordance with Dwg. 8799112.

5.5 Ammunition data cards. Data card information shall be as specified in MIL-STD-l167.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M230 and M231 Fuze provide safe arming and fuzing for the grenades used in the 2.75 Inch Rocket Multi-Purpose Submunition (MPSM).

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T04:27Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required. the specific issue of individual documents referenced (see 2.1. 1).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Certificate of conformance for each lot or shipment of product.

6.3 Consideration of Data requirements. The following data requirements should be considered when this specification is applied on a contract, per the Contract Data Requirements List (CDRL), DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
4.5.18	DI-NDTI-80809A	Test/Inspection Report	---

6.4 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC,, ATTN: SMCAR-QAF-A, Picatinny Arsenal, NJ. 07806-5000. This address will be specified on the CDRL, DD Form 1423 in the contract.

6.5 Drawing. Drawings listed in Section 2 of this specification under the heading U. S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM) or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 Submission of alternativequality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.7 Subject term (kev word) listing.

Hydra 70
2.75 Inch Rocket
Multi-purpose submunition
Ram air decelerator

6.8 Changes from the previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1340-A943)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-63446 (AR)	940912

3. DOCUMENT TITLE: FUZES, GENERAL PURPOSE GRENADE: M230 AND M231 (PRACTICE)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON	e. DATE SUBMITTED (YYMMDD)

7. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: AMSTA-AR-EDE-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:27Z
Check the source to verify that this is the current version before use.